

**ROBOTICS LAB**  
**IRO-2 SS25**  
**INDUSTRIAL DRIVES, PART 2**  
**Lab report (revised submission)**

**Group : 5.4**

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## TASK 1

In Task-1 we are given positions of a ball that was measured at 20 supporting points on the resulting paths from 10 experiments from a catapult system.

The trajectory of the path taken by the ball can be described as a quadratic equation:

$$z = ay^2 + by + c$$

Thus we have 200 pairs of y and z values (the z values are a function of y from the equation above) which leads to a system of 200 linear equations.

Since the number of equations are much greater than the number of unknowns ( $200 \gg 3$ ) and which also has some inaccuracies in the measurement, we wouldn't have a unique solution, but we can solve for the values of coefficients by least-squares method.

We can minimize the quadratic error by using the error formula on each equation by linear equation  $e = A * x - b$  which will result in a non-zero error vector which could further be used for finding the minimum error by summing up each error value's square, which in Matlab would be automatically done by using the \ operator.

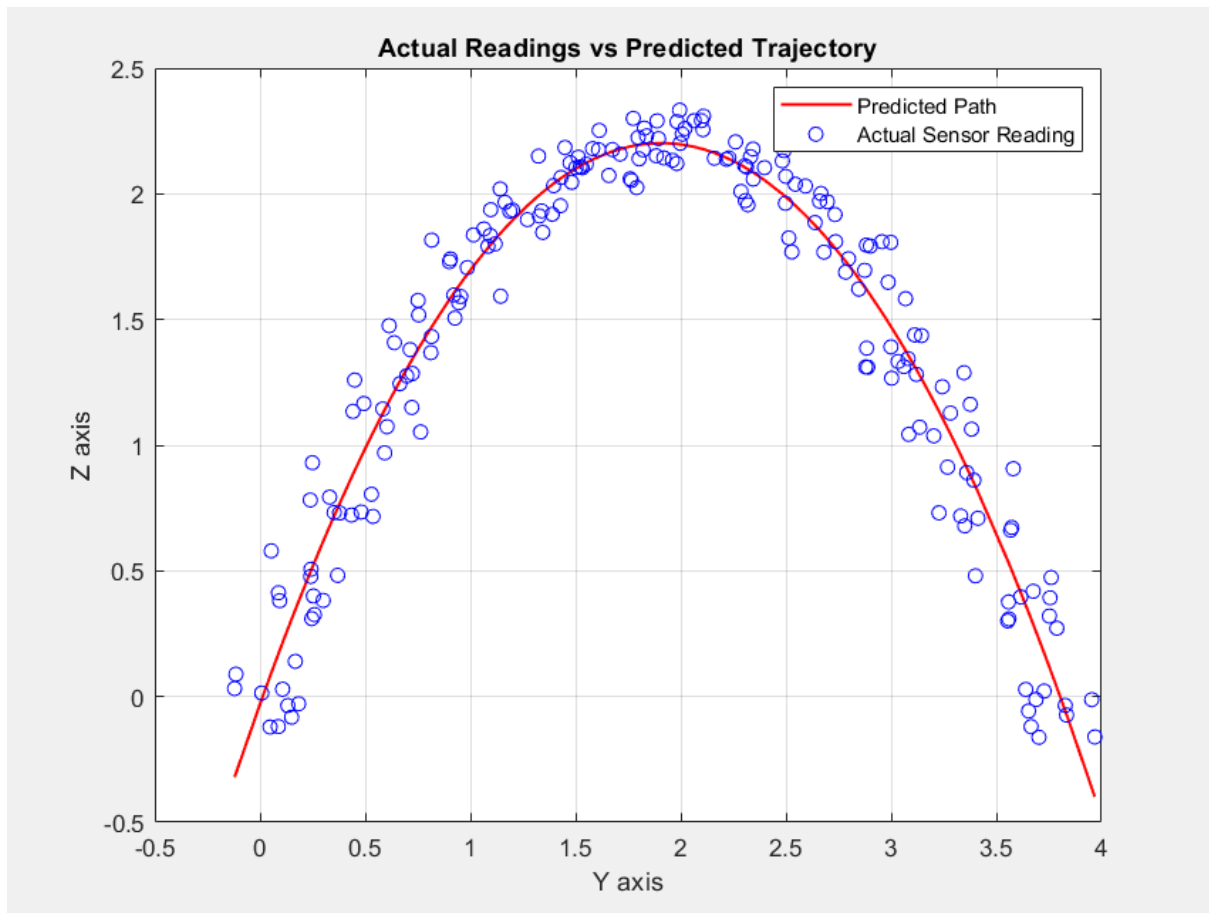
Therefore, we were able to find the coefficient values for the trajectory equations as :  $a = -0.6130$

$$\underline{b = 2.3391}$$

$$\underline{c = -0.0296}$$

With these coefficients we could use our trajectory equation for plotting the predicted path for our ball.

- The finalized function is  $z = -0.613y^2 + 2.3391y - 0.0296$



*Figure 1: Comparison of actual sensor reading and predicted path between Y and Z axes*

Here we could see the difference in the data, We used the actual sensor reading and minimized the error by least-squares sense and constructed the most plausible trajectory of our ball in respect to y axis and z axis

Remaining Quadratic Error : 6.6587 , Which is the sum of the square of vertical distances between the Actual reading and the quadratic curve. It tells us about how good of a fit of the curve that we are able to make from the actual sensor values.

## TASK 2

Analog signals require conversion into discrete-time signals through sampling.  $u(t)$  is an analog signal and its samples are  $u_k = u(kT)$ , where  $T$  is the sampling time, then discrete-time signal is  $u[k] = u_k$ .

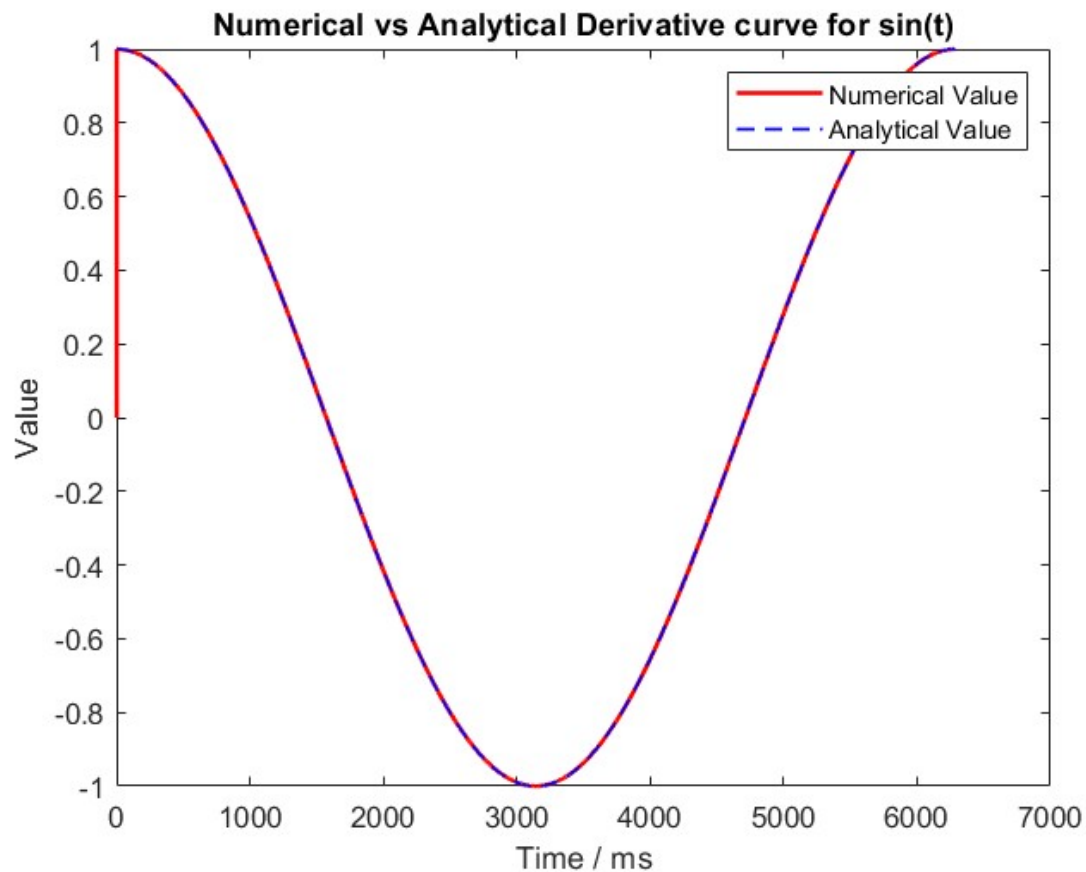
**Objective:** Implement a numerical discrete-time derivative function and verify its accuracy against an analytical solution. The numerical approximation for the time derivative  $\dot{u}[k]$  is given by:

$$\dot{u}[k] \approx \frac{u[k] - u[k-1]}{T}$$

**Implementation in MATLAB** function `num_derivative(u, T)` accepts a column vector  $u$  (discrete-time input signal) and a positive real number  $T$  (sampling time), returning a column vector  $y$  of the same length. The first value of  $y$  is set to zero as it cannot be determined.

We can verify if our function is working as intended by using simple mathematical function whose derivative is known.

- A MATLAB vector `usine = sin((0:0.001:2*pi)')`; (sampling time  $T=1$  ms (distance between 2 sampling instances)), and is the computed by `num_derivative` for numerical derivative.
- The analytical derivative of  $\sin(t)$  is  $\cos(t)$ , thus when we plot our derivative of our sine curve and we can see a cosine curve



*Figure 2: Comparison of numerical discrete time approximation and analytical true time derivative values*

The red line mentioned as Numerical Value is the plot using *num\_derivative* function and the blue dotted line mentioned as analytical value is the plot of cosine curve.

**Discussion:** The plot comparing the numerical and analytical solutions for the sine wave derivative showed exact the same, confirming that our differentiation method through discrete-time sampled representation is exactly the same.

## TASK 3

Servomotors, typically Permanent-Magnet Synchronous Motors (PMSMs), generate torque through the magnetic fields created by the stator windings and permanent magnets on the rotor. This torque is so called motor torque ( $M_{\text{Motor}}$ ), and is proportional to the torque-creating current component ( $I_q$ ), given by  $M_{\text{Motor}} = k_T \cdot I_q$

**Objective:** Calculate the torque constant  $k_T$  based on continuous and peak operating points and explain difference.

**Results:**

- Torque constant based on continuous stall values ( $M_0 = 1.4 \text{ Nm}$ ,  $I_0 = 1.8 \text{ A}$ ):  $k_{T0} = \frac{M_0}{I_0} = \frac{1.4 \text{ Nm}}{1.8 \text{ A}} = \underline{0.7778 \text{ Nm/A}}$
- Torque constant based on peak values ( $M_{\text{max}} = 3.5 \text{ Nm}$ ,  $I_{\text{max}} = 5.7 \text{ A}$ ):  $k_{T\text{max}} = \frac{M_{\text{max}}}{I_{\text{max}}} = \frac{3.5}{5.7} = \underline{0.6140 \text{ Nm/A}}$
- The difference between  $k_{T0}$  and  $k_{T\text{max}}$  is  $0.1637 \text{ Nm/A}$

**Explanation:** The difference between two values is attributed to magnetic saturation. At high current levels ( $I_q$ ), the iron parts within the motor begin to saturate and lose their ferromagnetic properties. This increases the magnetic resistance, so  $I_q$  increases lead to a disproportionally smaller increase in the magnetic field in the air gap, and thus in the motor torque. Therefore,  $k_T$  is not truly constant but decreases with increasing  $I_q$  at higher current levels.

## TASK 4

**Objective:** Determine the coefficients for a 2<sup>nd</sup> order polynomial approximation of  $k_T(I_q)$  and implement it in a MATLAB function.

The relationship between  $k_T$  and  $I_q$  is approximated by the quadratic equation:

$$k_T(I_q) \approx a \cdot I_q^2 + b \cdot I_q + c$$

The task is to solve the coefficients  $a$ ,  $b$  and  $c$ .

From  $k_T(I_0) = a \cdot I_0^3 + b \cdot I_0^2 + c \cdot I_0$  (equation 1), From  $k_T(I_{\max}) = a \cdot I_{\max}^3 + b \cdot I_{\max}^2 + c \cdot I_{\max}$  (equation 2) and  $\left. \frac{dk_T}{dI_q} \right|_{I_q=0} = 0$ ,

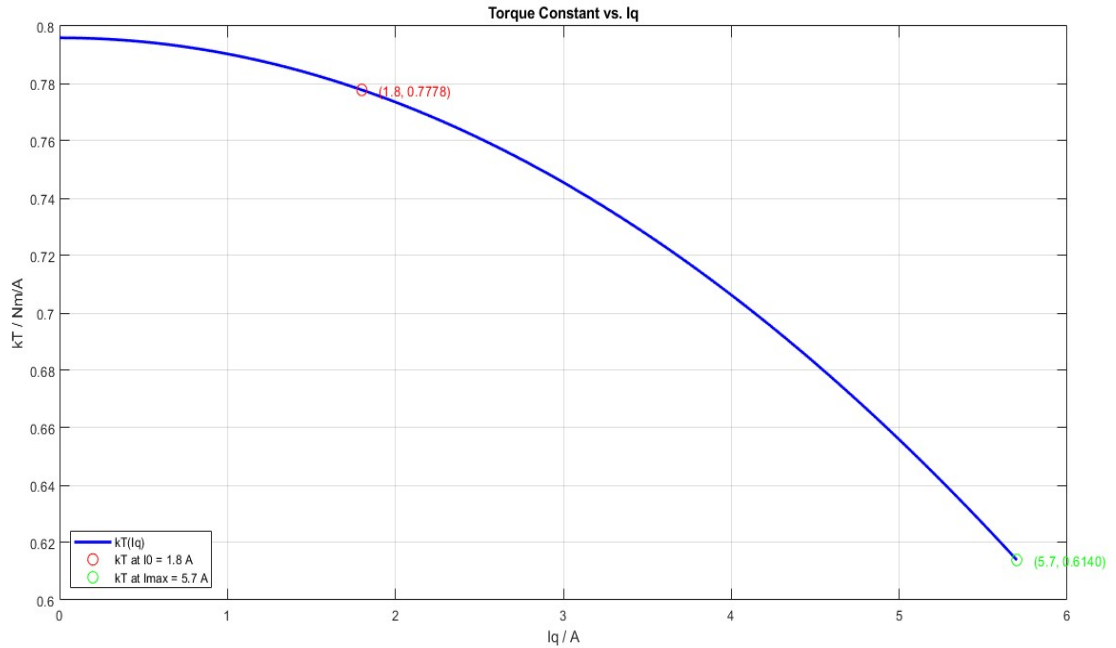
$$\frac{dk_T}{dI_q} = d(a \cdot I_q^2 + b \cdot I_q + c) / dI_q = 2aI_q + b = 0 \Rightarrow \underline{b = 0}.$$

and by using the \ operator in Matlab we were able to find the other coefficients:  $a = -0.0056$  and  $c = 0.7959$  by least-square method from the equation:  $M = k_T(I) \cdot I$ , by setting matrix  $I(\text{current}) = [I_0^3 \ I_0; I_{\max}^3 \ I_{\max}]$  and  $M(\text{Motor torque}) \text{ vector} = [M_0; M_{\max}]$ , so that  $k_T(I) = I \backslash M$

**Implementation:** A MATLAB function  $kT = \text{torqueconstant}(Iq)$  takes the current value of the current and substitutes the value in the quadratic equation. We can also conclude that the values of the coefficients are unique since the number of unknowns are same as the number of equations.

**Results:**

- The finalized relationship between  $k_T$  and  $I_q$  is  $k_T(I_q) = -0.0056 \cdot I_q^2 + 0.7959$
- The calculated  $k_{T0}$  (from polynomial at  $I_q = 1.8 \text{ A}$ ) is  $0.7778 \text{ Nm/A}$ .
- The calculated  $k_{T\max}$  (from polynomial at  $I_q = 5.7 \text{ A}$ ) is  $0.6140 \text{ Nm/A}$ .



**Figure 3: Plot of torque constant over current**

The result  $k_T(I_q)$  is monotonically decreasing, reflecting the impact of magnetic saturation at higher current levels. The polynomial model precisely matches the  $k_T$  values calculated in Task 3 for the continuous and peak operating points, verifying the accuracy.

## TASK 5

In the lab training systems, no gearbox and negligible additional load torque ( $M_{Load} = 0$ ), the equation simplifies to:  $M_{Motor} = J \cdot \alpha$ , where  $J$  (total moment of inertia)  $= J_{Load} + J_{Motor}$ . The task is to identify  $J_{Load}$ .

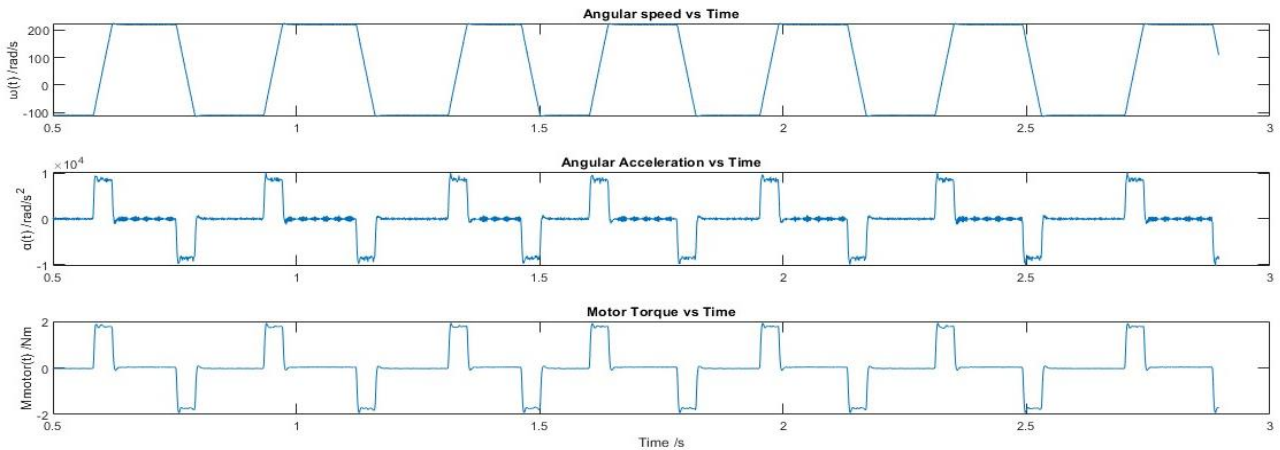
**Objective:** Import and process trace data from *JLoadWithoutFriction.trace* to prepare signals for inertia estimation.

**Data Import & Processing:**

- Import *JLoadWithoutFriction.trace* in function *load\_traces.m*, and extract angular speed  $\omega(t)$  and torque-creating current component  $I_q(t)$  we were able to find the right column indices with the help of 'names' array which corresponded to indices 3 and 4 respectively.



- Converted signals to SI units:  $[\omega(t)]$  from deg/s to rad/s, and  $[I_q(t)]$  from mA to A.
- Calculate angular acceleration  $\alpha(t)$  in  $[\text{rad/s}^2]$  by *num\_derivative* function.
- Calculate Motor torque  $M_{\text{Motor}}(t)$  in  $[\text{Nm}]$  by *torqueconstant* function.



*Figure 4: Plot of angular speed, angular acceleration and motor torque without friction over time*

## TASK 6

**Objective:** Estimate the total moment of inertia  $J$  from the simplified model and compare  $J_{\text{Load}}$  to theoretical values.

**Estimation of Total Moment of Inertia:**

- Calculate maximum absolute angular acceleration  $\alpha_{\text{max}}$ , and sample only where  $|\alpha(t)| \geq \alpha_{\text{max}}/4$  to avoid division by zero during constant speed phases.
- Calculate mean value of  $J_{\text{mean}}$  by MATLAB command mean from  $J = M_{\text{Motor}}/\alpha$

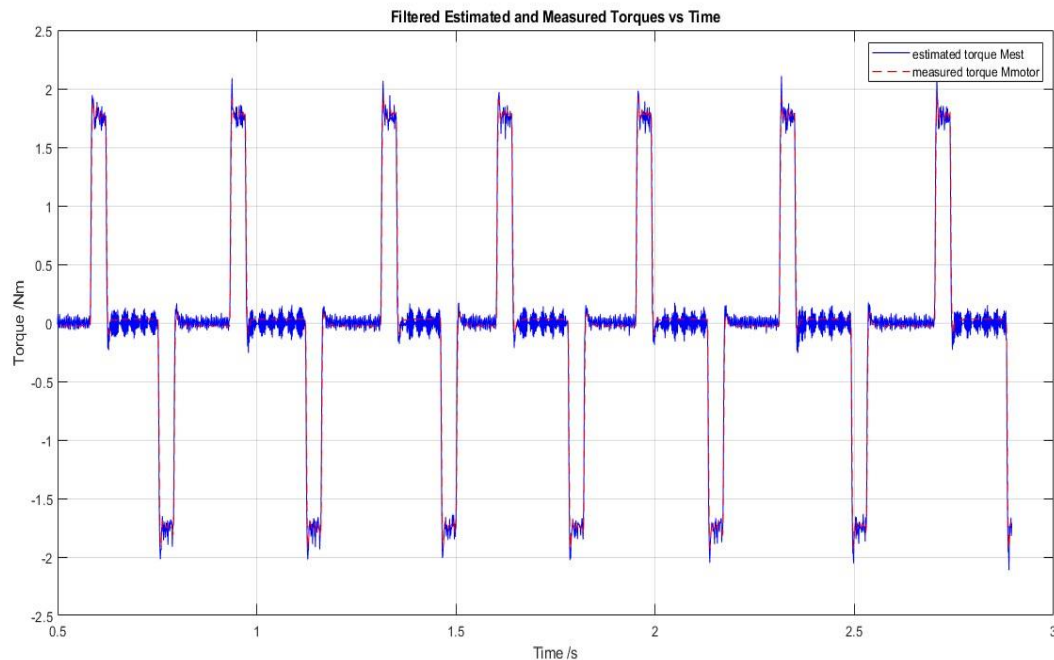
## Results:

- The estimated total moment of inertia  $J_{mean} = 2.0581 \times 10^{-4} \text{ kgcm}^2$ .
- Given  $J_{Motor} = 0.205 \text{ kgcm}^2$  (found from Industrial Drives 1) converted to  $0.205 \times 10^{-4} \text{ kgcm}^2$ , so  $J_{Load} = 1.8531 \text{ kgcm}^2$  from  $J_{Load} = J - J_{Motor}$ .

## Verification:

Compute estimated motor torque  $M_{Motor}^{est}(t) = J \cdot \alpha(t)$  for all samples.

The estimated  $J_{Load} = 1.808 \text{ kgcm}^2$  is approximately 25% higher than the theoretical value of  $J_{theo} = 1.445 \text{ kgcm}^2$  from preparation notes because the simplified model ( $M_{Motor} = J \cdot \alpha$ ) assumes that all motor torque is used for acceleration. In reality, some torque is always consumed by friction, so the inertia is overestimated as the friction torque is mistakenly attributed to acceleration and also unaccounted inertia by any small components (e.g., encoder) adding inertia that theoretical model doesn't include.

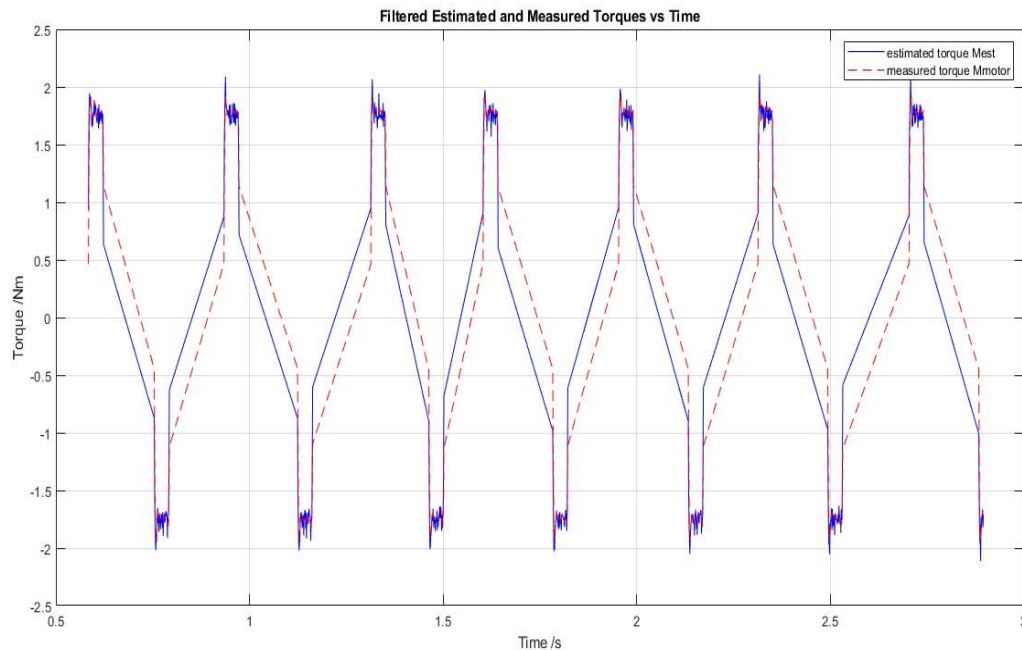


**Figure 5: Comparison of measured torque and estimated torque  $M_{est} = J \cdot \alpha$  over entire measurement time**

Measured motor torque is calculated from  $M_{Motor} = k_T(I_q) \cdot I_q$  representing actual torque generated by the motor based on  $I_q$ , while estimated motor torque is calculated using  $M_{est}(t) = J \cdot \alpha$ , estimating required torque to produce the observed angular acceleration without friction.

The graph is plotted to visually compare two torque signals across the entire time range and to validate the accuracy of the estimated inertia  $J$ , showing good match between the measured and estimated torques during high-acceleration periods indicates reliable inertia estimation.

Full-time plot of  $M_{est}(t) = J \cdot \alpha(t)$  includes segments where  $\alpha(t) \approx 0$ , leading to unreliable or noisy estimation due to near-zero division sensitivity. For more stable comparison, additional plot is included to focus only time intervals of  $|\alpha(t)| \geq \alpha_{max}/4$ . This filtered data aligns better behind the assumption  $M_{est} = J \cdot \alpha$ , and reflects more consistent estimation of dynamic system.



*Figure 6: Estimated VS. Measured torque during high-acceleration phase*

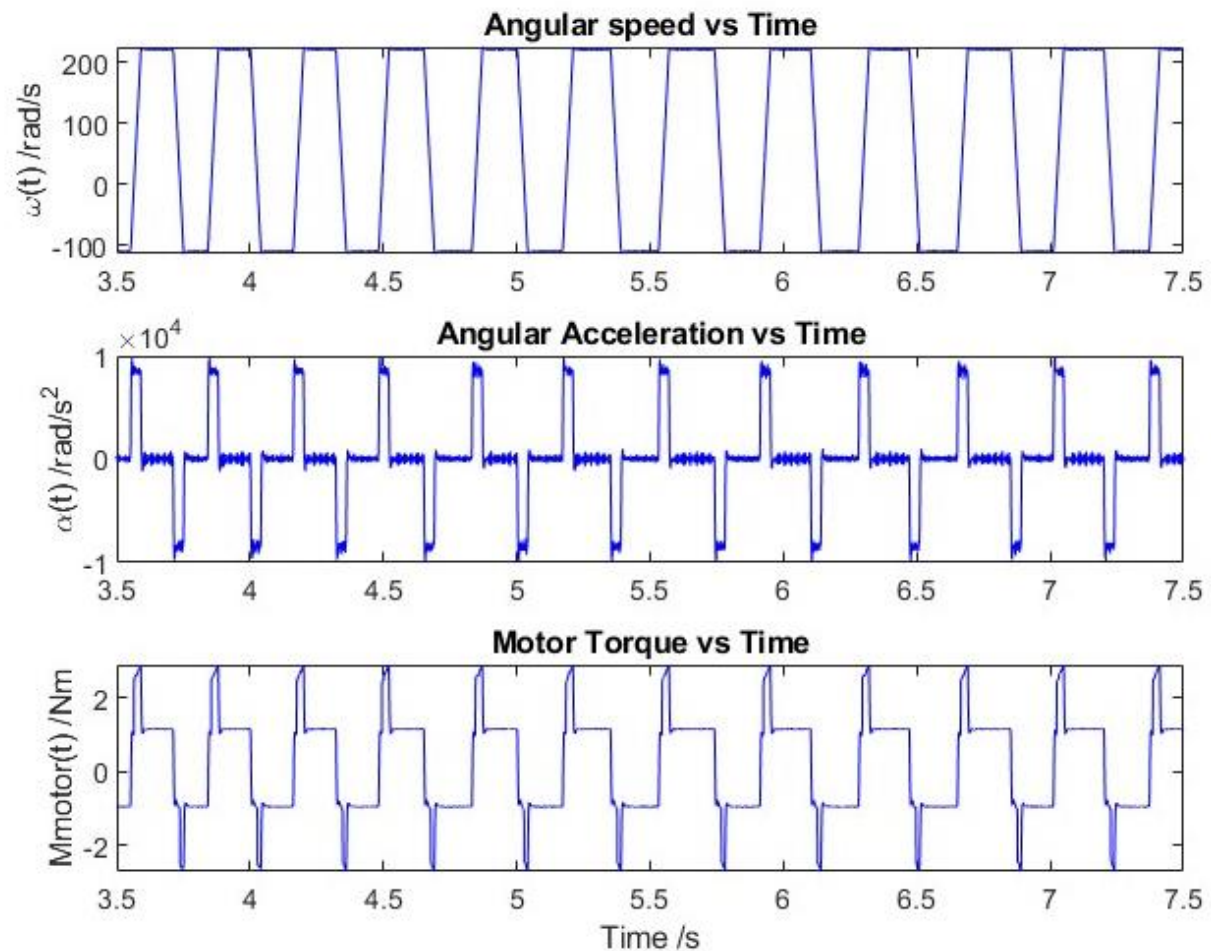
**Discussion:** This illustrates both the strength and the limitation of the simplified estimation as it diverges in low-acceleration phases due to neglected friction and noise in  $\alpha(t)$ .

## TASK 7

**Objective:** Import and process trace data from *JLoadAndFriction.trace* that includes friction.

**Data Import & Processing:**

- Import data from *JLoadAndFriction.trace*
- Convert  $[\omega(t)]$  and  $[I_q(t)]$  to SI units, and calculate  $\alpha(t)$  and  $M_{\text{motor}}(t)$  as in Task 5.



**Figure 7:** Plot of angular speed, angular acceleration and motor torque with friction over time

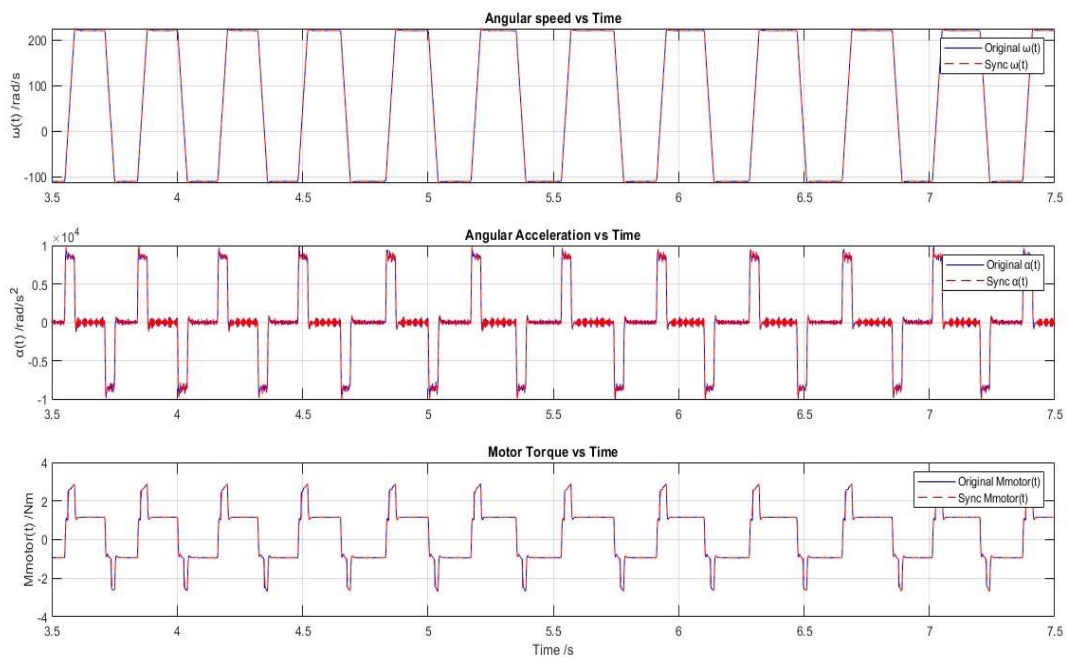
## TASK 8

**Objective:** Perform preliminary signal processing by removing samples at low angular speeds.

### Signal Filtering:

- Delete the samples where  $|\omega(t)| < 20 \text{ rad/s}$  to exclude the non-linear friction behaviour at very low speeds.
- Create a synchronized time vector  $t_{\text{sync}}$  for the remaining samples.

**Verification:** The plots confirm that samples corresponding to low speeds were successfully removed, allowing for more reliable friction modelling.



*Figure 8: Comparison between original and sync angular speed, angular acceleration and motor torque with friction over time*

**Angular Speed vs Time:** The original  $\omega(t)$  curve shows periods where the speed also includes zero, while sync  $\omega(t)$  curve demonstrates that removed low-speed segments, resulting in primary high-speed, dynamic phases of operation.

**Angular Acceleration vs Time:** Similarly, original  $\alpha(t)$  contains noise and artifacts around zero acceleration, especially in low speed. Sync  $\alpha(t)$  signal is cleaner as we discard data points corresponding to problematic low-speed friction.

**Motor Torques vs Time:** The plot shows original  $M_{\text{motor}(t)}$  and sync  $M_{\text{motor}(t)}$ . It confirms the consistency of the remaining data and the visual agreement between the original and synchronized torque, which is crucial for the subsequent parameter estimation. Filtering out of low speed makes the data more suitable for the linear friction model used in later tasks.

The synchronization process is critical because the linear friction model doesn't accurately represent friction at very low speeds or during direction changes. By excluding these problematic data points, the remaining dataset is more consistent with the linear model, thus enabling more accurate least-squares estimation of the mechanical parameters ( $J$ ,  $K_{\text{Fric},\omega}$ , and  $K_{\text{Fric},0}$ ) on the next task. The visual comparison confirms that removal of undesired data segments doesn't create distortion on the overall dynamic behaviour of the drive system.

## TASK 9

**Objective:** Set up and solve the overdetermined system of equations using the least-squares method to estimate  $J$ ,  $K_{\text{Fric},\omega}$  and  $K_{\text{Fric},0}$ .

Equation of motion for a drive axis is given by:

$$M_{\text{Motor}} = J \cdot \alpha + K_{\text{Fric},\omega} \cdot \omega + K_{\text{Fric},0} \cdot \text{sgn}(\omega) + M_{\text{Load}}$$

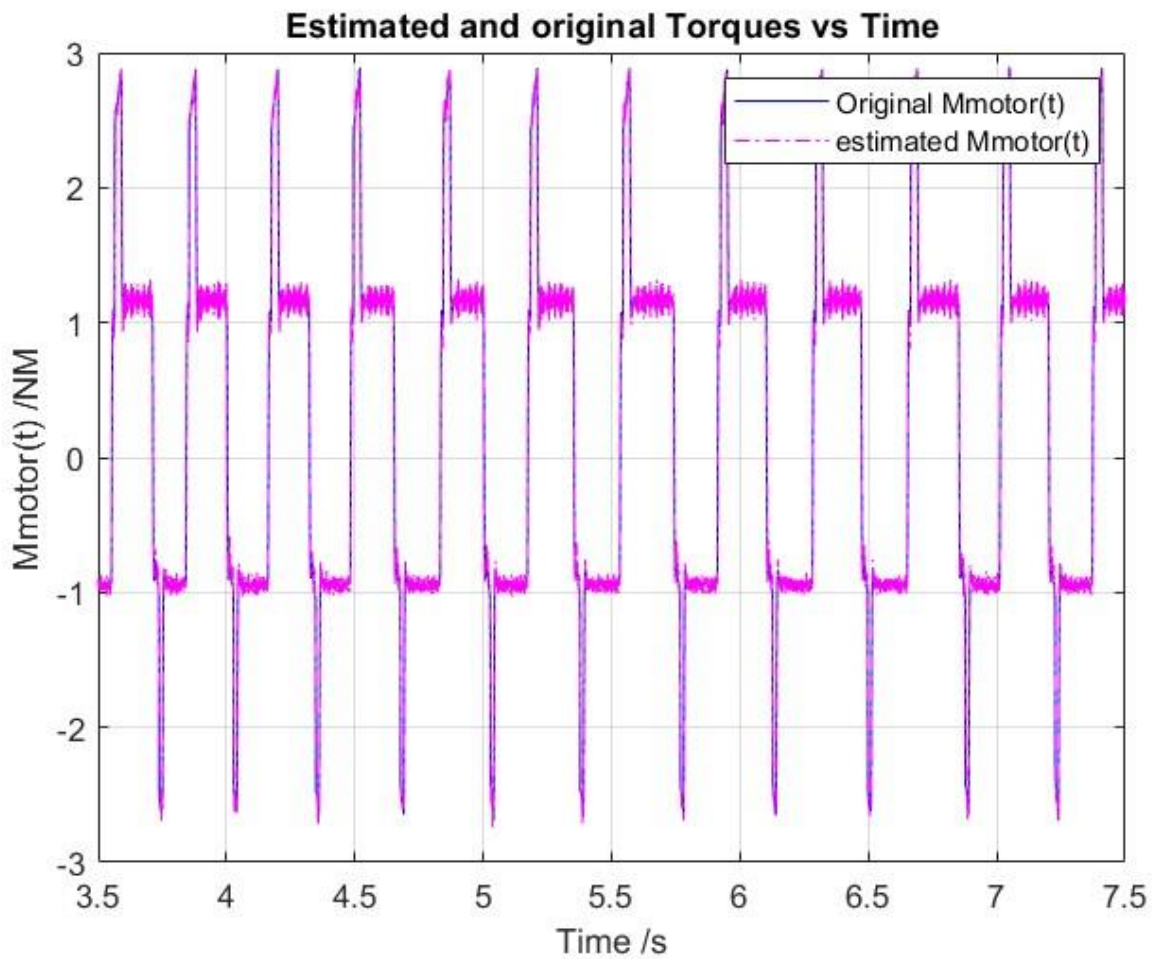
**Least-Squares Identification:**

The equation  $M_{\text{Motor}} = J \cdot \alpha + K_{\text{Fric},\omega} \cdot \omega + K_{\text{Fric},0} \cdot \text{sgn}(\omega)$  was rearranged into the linear system form, so  $X = [J; K_{\text{Fric},\omega}; K_{\text{Fric},0}]$ , and construct matrix  $A$  that such that it represents  $[\alpha_{\text{filt}}(t_{\text{sync},n}), \omega(t_{\text{sync},n}), \text{sgn}(\omega(t_{\text{sync},n}))]$  then construct vector  $M_{\text{sync\_est}} = M_{\text{Motor}}(t_{\text{sync},n})$  (using the time indices we filtered out) And we would be solving the system by using the  $\backslash$  operator ( $A \backslash M_{\text{sync\_est}}$ ) and get the least-squares solution for each of the coefficients.



## Results:

- The estimated mechanical parameters are: Total moment of inertia  $J = 2.0080 \times 10^{-4} \text{ kgm}^2$ , Viscous friction coefficient  $K_{Fric,\omega} = 0.0020 \text{ Nm/rads}$ , and Coulomb friction  $K_{Fric,0} = 0.7278 \text{ Nm}$ .
- The corresponding quadratic error  $E = 52.3462$ .
- The  $J_{Load}$  based on this identified  $J$  is  $1.803 \text{ kgcm}^2$  is calculated from  $J_{Load} = J - J_{Motor}$  where  $J_{Motor} = 0.205 \text{ kgcm}^2$  (converted to  $0.205 \times 10^{-4} \text{ kgm}^2$ )



*Figure 9: Comparison between original and estimated motor torque with friction over time*

**Discussion:** Original  $M_{\text{Motor}}(t)$  represents measured motor torque, while estimated  $M_{\text{sync\_est}}(t)$  represents predicted torque from the equation:  $M_{\text{Motor}} = J \cdot \alpha + K_{\text{Fric},\omega} \cdot \omega + K_{\text{Fric},0} \cdot \text{sgn}(\omega)$ . The parameters  $J$ ,  $K_{\text{Fric},\omega}$ ,  $K_{\text{Fric},0}$  are derived from least-squares fit on the synchronized speed data. The estimated  $J_{\text{Load}} = 1.803 \text{ kgcm}^2$  from friction model is closer to the theoretical value of  $J_{\text{theo}} = 1.445 \text{ kgcm}^2$  than from without friction ( $1.8531 \text{ kgcm}^2$  in Task 6). This highlights the importance of friction in the model by accounting for the torque consumed by friction, the least-squares method more accurately attributes the remaining torque to acceleration.

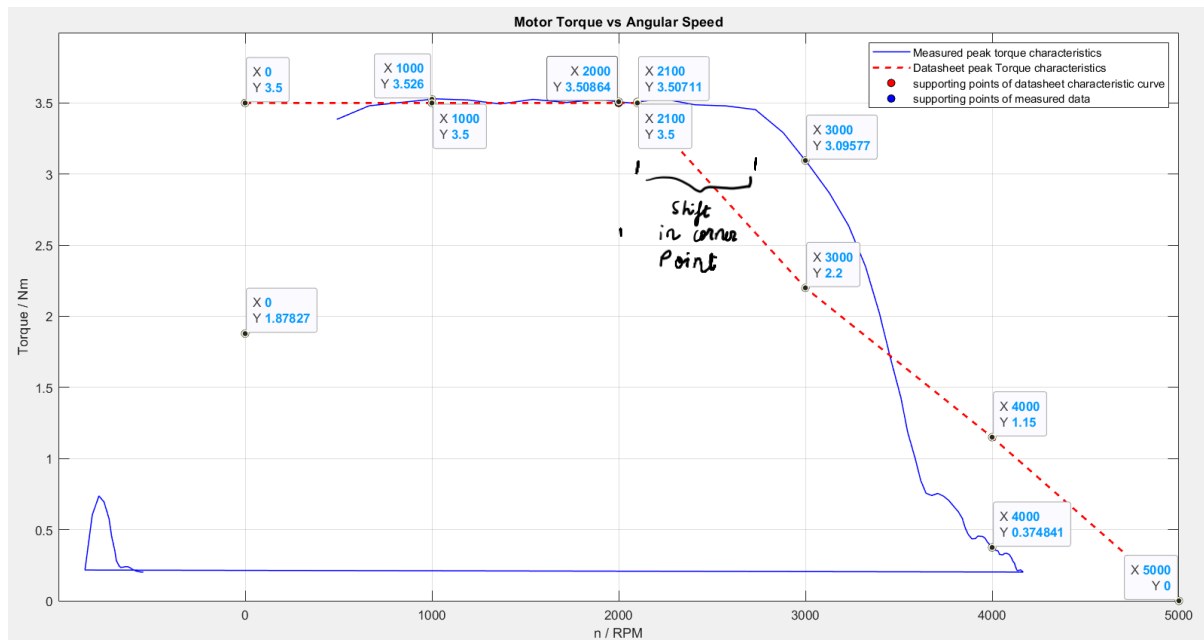
## TASK 10

**Objective:** Generate a speed-torque curve from measured data and compare it to the datasheet limit.

**Data Import, processing, extraction and plotting:**

- Import data from *MaxAcc.trace*, and convert current  $I_q(t)$  to [A] and angular speed  $\omega(t)$  to rotational speed  $n(t)$  in [rpm].
- Calculate Motor torque  $M_{\text{motor}}(t)$  in [Nm] by *torqueconstant* function.
- Remove samples of which  $M_{\text{motor}}(t) < 0.2 \text{ Nm}$ , and focus only on high-torque operating range by deleting all samples prior to the first  $M_{\text{motor}}(t) > 3.0 \text{ Nm}$ .
- Plotting the remaining samples that we have. (after excluding)





**Figure 10: Comparison of peak motor torque from datasheet and measured peak motor torque over angular speed ( $n$ )**

| $n$ [rpm]        | 0    | 1000 | 2000 | 2100 | 3000 | 4000 | 5000 |
|------------------|------|------|------|------|------|------|------|
| $M_{max}$ [Nm]   | 3.5  | 3.5  | 3.5  | 3.5  | 2.2  | 1.15 | 0    |
| $M_{motor}$ / Nm | 1.88 | 3.53 | 3.51 | 3.51 | 3.1  | 0.37 | 0    |

**Figure 11: Supporting point values that we used to plot the characteristic curve of the peak motor torque from the datasheet and points interpolated on the measured peak torque characteristics**

## TASK 11

**Objective is to Explain differences between the measured and datasheet speed-torque characteristics, we can see there is a difference in the corner points from the previous task.**

**Explanation:** The general shapes of both curves are similar, but there are notable differences, particularly in the corner point where the peak torque starts to decline. These differences are from two main reasons:

## 1. Temperature Dependence of Winding Resistance:

- **Datasheet Assumption:** Datasheet is made for the motor winding that are tested at 120°C.
- **Lab Conditions:** the motor was at approximately 20°C room temperature and not pre-heated.
- **Impact:** The resistance of copper windings increases with temperature. Therefore, at 20°C, the winding resistance was considerably lower (approximately 70%) than the value assumed for in datasheet. A lower resistance means that for given supply voltage, a higher current and torque are driven into the motor and allow the motor to maintain peak torque up to higher rotational speeds which contributed slightly to the shift in the “knee point” (corner point)

## 2. Dynamic vs. Stationary Conditions (Inductance Effect):

- **Datasheet Characteristic:** In datasheet curve, peak torque is under stationary conditions, modelled by a simplified equivalent circuit (Figure 5) that neglects inductance.
- **Measured Characteristic:** The lab measurement was performed during a high dynamic acceleration process. The inductance (L) of the motor winding (as shown in Figure 7) becomes significant.
- **Impact:** The inductor introduces a voltage drop that opposes changes in current. During rapid acceleration, portion of supply voltage is consumed by this inductive voltage drop, so it limits the voltage available to drive the current component  $I_q$  that creates torque. Since the inductive voltage drop opposes rapid change in current, this would enable a extended current delivery for a very short period of time which leads to a slight delay until the inductive back-EMF is fully dominating the supply voltage. This effect is also contributed by Advanced controllers which can push more current using the voltages that are available in the bus topologies pushing even more speed.

This results in the observed phenomenon where the measured peak torque starts to decline at a higher speed compared to the datasheet's stationary model.